

Track 2

3-Day Dataset Validation Sprint

Team assignment and execution plan

Approach

Work is split by dataset track and source complexity, with one team member assigned as Pipeline Integrator. Splitting by track keeps ownership clear and avoids two failure modes: everyone validating the same data (conflicting results) and everyone working in isolation on unrelated tasks (bottlenecks at handoff).

1. Dataset and Role Assignment

Shamsudeen Bankole— Classification (High-Volume / Public Data)

- **Datasets:** All Kaggle datasets (Papers 10, 11, 12, 13, 14, 15, 16, 17).
- **Focus:** 2D MRI slices, high volume, simpler labels.
- **Primary task:** Map folder names to the four approved classes (glioma, meningioma, pituitary, no_tumor). Check for exact duplicates — Kaggle datasets frequently contain duplicated slices.

Md Tasnimul Hoque— Classification & Clinical (Complex / Institutional Data)

- **Datasets:** TCIA/TCGA collections (Papers 2, 4, 5), Swedish Pediatric (Paper 36), and smaller institutional classification datasets (Papers 26, 30, 35, 45, 46, 47, 49–53, 56, 58–61).
- **Focus:** 3D volumes or complex clinical metadata, ambiguous labels.
- **Primary task:** Heavy label mapping (e.g. mapping Glioblastoma, GBM, and Astrocytoma all to glioma). Verify patient IDs to prevent leakage across these smaller, messy datasets.

Naman Malik— Segmentation (BraTS)

- **Datasets:** All BraTS datasets (Papers 1, 3, 6, 7, 8, 9, 18, 19, 20, 21, 22, 44, 66, 67).
- **Focus:** 3D NIfTI volumes, multi-modal (T1, T1ce, T2, FLAIR), multi-region masks.
- **Primary task:** Validate BraTS 2020 and strictly enforce the frozen 295/74 split. Validate mask geometries, ensure no cross-split patient leakage, and tag mask types (BraTS_WT_TC_ET vs. binary). This is the critical-path task for the sprint.

Harsh— Segmentation & Pipeline Integrator (Non-BraTS + Automation)

- **Datasets:** Non-BraTS segmentation datasets (Papers 41, 42, 43, 57, 62, 63, 65), plus team automation duties.
- **Focus:** Metastasis, meningioma, and institutional 3D segmentation data.
- **Primary task:** Validate assigned datasets and act as technical : write and run the scripts that merge everyone's work, generate the symlinks for the model-ready directory, and compile the final reports.

2. Deliverables

The team provides one unified template on Day 1 morning:

- **Master manifest template** — A blank CSV with the agreed headers (sample_id, subject_id, track, final_class_label, validation_status, duplicate_group_id, mask_status, final_split, source_dataset, modality, image_path, mask_path).
- **Assignment list** — A text file listing exactly which datasets each member owns.
- **Rules document** — The 3-day sprint constraints: no resizing, BraTS 2020 frozen, patient-level splitting.

Deliverable	Submitted by	Submission method
Validated master manifest	All	Each member completes their assigned rows in master_manifest_template.csv and submits their individual file by Day 2, end of day.
Exception / rejection log (Optional)	All	No separate file is created. Members filter their validation_status column for REJECTED or REVIEW and send those rows.
Quick validation report (Optional)	All	Each member writes a one-paragraph summary of their track (e.g. duplicate rate, label mapping issues) and sends it to Member 4.
Dataset analysis summary (Optional)	Optional	Member 4 runs a script on the merged manifest to generate analysis_summary.csv.
Model-ready directory (Optional)	Optional	Member 4 runs the symlink script on the final merged manifest to build the /model_ready/ folder structure.